OT-Hybrid-BO: Optimal Transport-Balanced LLM–GP Bayesian Optimization for Hyperparameter Tuning

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Abstract

Bayesian Optimization (BO) is the standard approach for hyperparameter tuning of expensive machine learning models. Classical Gaussian Process (GP) surrogates suffer from a cold-start problem with sparse observations, while LLAMBO (**?**), which replaces the GP with a Large Language Model (LLM), incurs prohibitive inference cost at every iteration. We propose **OT-Hybrid-BO**, a framework that maintains both an LLM-based candidate distribution and a GP acquisition distribution in parallel, using the Wasserstein-2 distance between them as a data-driven signal to dynamically arbitrate their relative influence at each iteration. When data is scarce and the GP is unreliable, the LLM leads; as observations accumulate and the GP matures, it progressively takes over. We show that this OT-based arbitration acts as an implicit GP misspecification detector, shifting trust toward the LLM precisely when GP assumptions are violated. Evaluated on HPOBench, OT-Hybrid-BO matches LLAMBO's sample efficiency in the cold-start regime while reducing total LLM inference cost and converging to GP-level performance as the budget grows.

1 Motivation

Hyperparameter optimization (HPO) is a central bottleneck in deploying machine learning models: identifying well-performing configurations of learning rate, batch size, regularization, and architecture choices requires evaluating expensive objectives such as validation loss after full training runs. Bayesian Optimization addresses this by maintaining a probabilistic surrogate of the objective f and using an acquisition function to select the next configuration to evaluate.

The dominant surrogate choice is the Gaussian Process (GP), which offers calibrated uncertainty and strong sample efficiency in smooth, low-dimensional search spaces (**?**). However, the GP has two well-known failure modes relevant to HPO. First, the *cold-start problem*: with fewer than approximately 10–15 observations, the GP has insufficient data to estimate a useful posterior, and its acquisition surface is nearly uninformative. Second, *model misspecification*: the GP assumes a stationary, smooth landscape via its kernel; when the true objective is discontinuous (e.g., across categorical choices) or highly multimodal, the surrogate is systematically wrong regardless of how much data accumulates.

LLAMBO (**?**) addresses the cold-start problem by replacing the GP with an LLM prompted with the task description and observed history to propose and evaluate configurations in-context. Because the LLM encodes rich semantic prior knowledge from pretraining, it generates high-quality candidates from the very first iteration. However, LLAMBO has a complementary weakness: it invokes the LLM at *every* iteration regardless of how much data has been collected. A recent reproducibility study (**?**) confirmed that LLAMBO's surrogate is weaker than GP or SMAC as a pure single-task regressor once sufficient data exists, while retaining strong advantages in the early, sparse regime.

This suggests the LLM and GP have *complementary* strengths: the LLM is most valuable early, the GP most valuable later.

A concurrent line of work, LLM-AutoOpt (**?**), proposes a hybrid HPO framework combining BO with LLM-based reasoning, but uses a hand-designed, fixed balance between the two agents. **The key missing ingredient is a principled, data-driven mechanism to arbitrate between the LLM and GP at each iteration.**

We propose that Optimal Transport (OT) provides exactly this mechanism. Specifically, the Wasserstein-2 (W_2) distance between the LLM's candidate distribution and the GP's acquisition distribution measures their *epistemic disagreement* at each step. We show that this distance is large precisely when the two agents disagree— which occurs both in the cold-start regime (GP is flat and uninformative) and when the GP is misspecified (GP concentrates on wrong regions). In both cases, the W_2-derived weight shifts influence toward the LLM. As the GP matures and its acquisition aligns with the LLM's semantic prior, W_2 shrinks, and the GP takes over. The misspecification-detection behavior emerges from the OT geometry rather than being explicitly programmed.

2 Related Work

LLAMBO. Liu et al. (**?**) demonstrate that LLMs, prompted in-context with observed hyperparameter–performance pairs, can warm-start and guide BO without fine-tuning. LLAMBO significantly improves early-iteration regret on Bayesmark and HPOBench tasks. Our work builds directly on LLAMBO's LLM arm but introduces a GP surrogate and OT-based balance to reduce the per-iteration LLM cost and improve late-stage performance.

Reproducibility and limitations of LLAMBO. Rychert et al. (**?**) reproduce LLAMBO with open-weight Llama 3.1 70B and confirm its warmstarting gains while quantifying its weakness as a pure surrogate model in data-rich regimes. Their finding directly motivates our hybrid design: the LLM is valuable early; the GP is valuable late.

Pre-trained GP priors for BO. HyperBO (**?**) pre-trains GP hyperparameters on historical HPO data, showing that informative priors improve BO efficiency substantially. HyperBO requires historical task data; our method uses LLM knowledge as a zero-shot prior and does not require prior runs on related tasks.

Hybrid LLM–BO frameworks. LLM-AutoOpt (**?**) combines BO with LLM meta-knowledge for time-series HPO, achieving improved predictive performance and interpretability. Unlike LLM-AutoOpt, which uses a fixed LLM–BO balance, our method derives the balance dynamically from the W_2 distance between the two agents' distributions.

Optimal transport for Bayesian inference. Wang et al. (**?**) show that OT provides tractable mechanisms for working with Bayesian posterior distributions, supporting the technical feasibility of our approach. We leverage the closed-form W_2 distance between Gaussians, making OT integration computationally negligible relative to the dominant GP and LLM costs.

3 Proposed Method: OT-Hybrid-BO

3.1 Setup

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the hyperparameter search space and $f : \mathcal{X} \to \mathbb{R}$ the black-box objective (e.g., validation loss after training). At iteration t, we have a dataset $\mathcal{D}_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^t$ of evaluated configurations. The goal is to find $\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{X}} f(\mathbf{x})$ with as few evaluations as possible.

3.2 Two parallel surrogates

LLM arm. Following LLAMBO, we serialize $\mathcal{D}_t$ and a task description into a structured prompt and query the LLM to propose k candidate configurations. We fit a multivariate Gaussian $P_{\text{LLM}}^{(t)} =$

$\mathcal{N}(\boldsymbol{\mu}_L, \boldsymbol{\Sigma}_L)$ to these k samples. This distribution encodes the LLM's semantic belief about promising regions of $\mathcal{X}$.

GP arm. We fit a GP surrogate $\mathcal{GP}(\mu_{\text{GP}}, k)$ on $\mathcal{D}_t$ and evaluate the Expected Improvement (EI) acquisition function on a candidate grid $\mathcal{G} \subset \mathcal{X}$. Normalizing EI over $\mathcal{G}$ yields a discrete distribution $P_{\text{GP}}^{(t)}$, which we approximate as a Gaussian $\mathcal{N}(\boldsymbol{\mu}_G, \boldsymbol{\Sigma}_G)$ for the OT computation.

3.3 OT-based arbitration

The Wasserstein-2 distance between two multivariate Gaussians has a closed-form expression:

$$W_2^2(P_L, P_G) = \|\boldsymbol{\mu}_L - \boldsymbol{\mu}_G\|^2 + \operatorname{Tr}\left(\boldsymbol{\Sigma}_L + \boldsymbol{\Sigma}_G - 2\left(\boldsymbol{\Sigma}_L^{1/2}\boldsymbol{\Sigma}_G\boldsymbol{\Sigma}_L^{1/2}\right)^{1/2}\right). \tag{1}$$

We derive a blending weight from this distance:

$$\lambda_t = \sigma\left(-\beta \cdot W_2(P_L^{(t)}, P_G^{(t)}) + \gamma/t\right), \tag{2}$$

where σ is the sigmoid function, $\beta > 0$ controls sensitivity to disagreement, and $\gamma > 0$ controls LLM influence decay over iterations. The blended acquisition is:

$$A(\mathbf{x}) = \lambda_t \cdot P_{\text{LLM}}(\mathbf{x}) + (1 - \lambda_t) \cdot P_{\text{GP}}(\mathbf{x}), \qquad \mathbf{x}^* = \arg\max_{\mathbf{x} \in \mathcal{G}} A(\mathbf{x}). \tag{3}$$

Why W_2 acts as a misspecification detector. When the GP is well-fitted, its acquisition concentrates near the same regions the LLM favors, yielding small W_2 and low λ_t. When the GP is misspecified or data-starved, its acquisition is diffuse or misdirected; the LLM's distribution remains concentrated in semantically plausible regions; W_2 grows large; and λ_t rises. The system shifts trust toward the LLM precisely when the GP is least reliable—without explicit programming.

Computational cost. Equation (**??**) requires only matrix operations on $d \times d$ covariance matrices, incurring $O(d^3)$ cost per iteration independent of $|\mathcal{G}|$. For higher-dimensional spaces where the Gaussian approximation is inadequate, we use Sliced Wasserstein distance, which is tractable via random one-dimensional projections.

3.4 Algorithm summary

1. Initialize: prompt LLM for k_0 warm-start configurations; evaluate f; set $\mathcal{D}_0$.
2. For $t = 1, 2, \ldots, T$:
   (a) Fit GP on $\mathcal{D}_{t-1}$; compute $P_{\text{GP}}^{(t)}$ via EI over $\mathcal{G}$.
   (b) Query LLM with $\mathcal{D}_{t-1}$ and task description; fit $P_{\text{LLM}}^{(t)}$.
   (c) Compute $W_2(P_L^{(t)}, P_G^{(t)})$ via Eq. (**??**); derive λ_t via Eq. (**??**).
   (d) Select $\mathbf{x}_t = \arg\max A(\mathbf{x})$ via Eq. (**??**).
   (e) Evaluate $y_t = f(\mathbf{x}_t)$; append to $\mathcal{D}_t$.

4 Feasibility and 7-Week Plan

4.1 Starting point and codebase

OT-Hybrid-BO extends the open-source LLAMBO repository[1], which provides the full LLM prompting pipeline, surrogate model implementations, and HPO benchmark integration (Bayesmark, HPOBench). The required extensions are: (a) adding a GP surrogate via BoTorch/GPyTorch (well-documented, ∼50 lines); (b) computing W_2 between two fitted Gaussians (closed-form, ∼15 lines of NumPy); and (c) implementing the blended acquisition (Eq. (**??**), ∼20 lines). No novel ML infrastructure is required. The LLM backbone will be Llama 3.1 70B (open-weight), validated by Rychert et al. (**?**) to reproduce LLAMBO's GPT-3.5 results.

[1] `https://github.com/tennisonliu/LLAMBO`

4.2 Timeline

Table 1: Seven-week implementation and evaluation plan.

Week	Task	Deliverable
1	Read LLAMBO codebase; reproduce baseline results on Bayesmark subset (5 tasks)	Confirmed baseline numbers
2	Implement GP surrogate arm (BoTorch EI) alongside LLAMBO pipeline	Working GP-BO in LLAMBO framework
3	Implement W_2 module (Gaussian closed-form + Sliced Wasserstein variant); unit-test against known distributions	Verified OT module
4	Implement blended acquisition (Eq. **??**); full OT-Hybrid-BO loop; sanity-check on 2D synthetic functions	End-to-end system running
5	Main experiments: OT-Hybrid-BO vs. LLAMBO vs. GP-BO vs. Random on HPOBench ($\geq$5 model–dataset pairs)	Draft results tables
6	Ablations: fixed λ vs. OT-driven; W_2 correlation with GP error; sensitivity to β, γ	Ablation figures + diagnostic plots
7	Write-up; finalise figures; prepare submission	Final report

4.3 Risks and mitigations

Risk 1: W_2 signal is noisy in high dimensions. HPOBench tasks are low-dimensional (2–9 hyperparameters), where Gaussian approximations of both distributions are reliable. For higher-dimensional tasks, Sliced Wasserstein provides a stable, scalable alternative.

Risk 2: LLM inference cost during experiments. Using Llama 3.1 70B locally eliminates API cost. Rychert et al. (**?**) confirm this backbone reproduces GPT-3.5 LLAMBO results faithfully.

Risk 3: W_2 does not correlate with GP reliability in practice. This is itself a falsifiable empirical question. If W_2 does not track GP error, we report this finding explicitly and fall back to a fixed-decay schedule for λ_t, which still constitutes a principled hybrid BO contribution. The core method is valid regardless of whether OT provides the optimal arbitration criterion.

5 Expected Contributions and Evaluation

Primary claim. OT-Hybrid-BO achieves competitive simple regret with LLAMBO in the early (data-sparse) regime while matching or surpassing GP-BO in the late (data-rich) regime, using fewer total LLM calls than LLAMBO across the full budget.

Secondary claims.

- The W_2 distance correlates with GP predictive error, validating its role as a misspecification-detection signal.
- The OT-derived λ_t outperforms a fixed λ in an ablation study.
- The method is robust to the choice of LLM backbone (Llama 3.1 70B vs. smaller models).

Evaluation protocol. We follow the Bayesmark and HPOBench evaluation protocol of Liu et al. (**?**) and Rychert et al. (**?**): normalized simple regret over $T = 25$ iterations, averaged over 5 random seeds, on tasks from the Random Forest, SVM, MLP, AdaBoost, and Logistic Regression model classes across at least five datasets.

Target venue. A clean empirical paper with the above results is targeted at an AutoML workshop (NeurIPS/ICML 2026) or the AutoML conference. A positive result on the W_2 misspecification-detection claim, supported by a formal lemma bounding the W_2 distance as a function of GP kernel error, would justify a main track submission.

References

References

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